



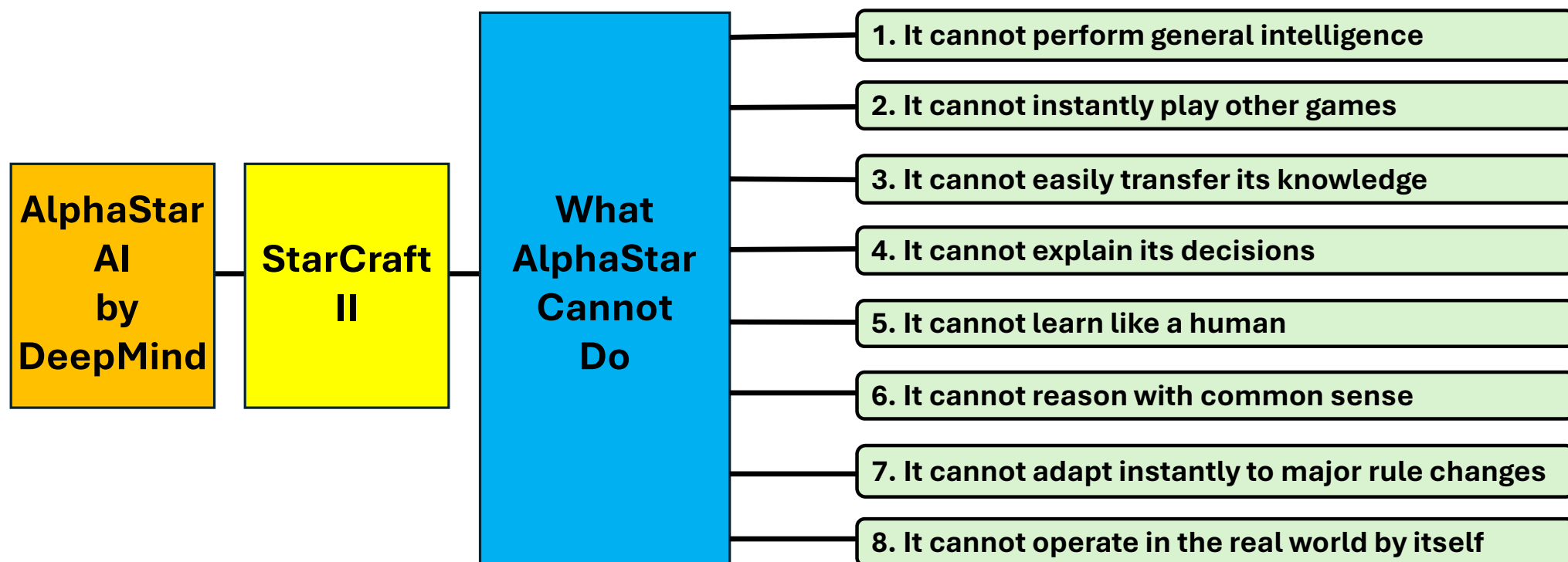
Team Tiger

ITAI 1370 Artificial Intelligence History

A03 What AlphaStar Cannot Do

Version 2.0

AlphaStar is an AI system developed by DeepMind to play the real-time strategy game StarCraft II at a superhuman level. Although it is extremely powerful within that game, it has many important limitations.



What AlphaStar Cannot Do: Limitations of Narrow AI

1. It cannot perform general intelligence

AlphaStar is designed for one specific task: playing StarCraft II. It cannot:

- Drive cars
- Carry on natural conversations
- Write essays
- Solve mathematical problems outside its training
- and Cook or perform everyday human tasks

It is an example of **Narrow AI**, not Artificial General Intelligence (AGI).

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2. It cannot instantly play other games

Even though AlphaStar mastered StarCraft II, it cannot automatically play:

- Chess
- Poker
- and Minecraft.

Each new game requires substantial retraining.

3. It cannot easily transfer its knowledge

The strategies AlphaStar learned in StarCraft II do not automatically apply to:

Business planning
Military operations
Robotics
and Medical diagnosis

Humans can often transfer knowledge between different domains; AlphaStar cannot.

4. It cannot explain its decisions

AlphaStar may make excellent strategic decisions, but it generally cannot explain:

- Why it chose a particular attack
- Why it expanded its base
- Why it selected one strategy over another

This lack of interpretability is common in deep neural networks.

5. It cannot learn like a human

Humans can learn a new game after reading the rules or watching a few matches.

AlphaStar required:

- Hundreds of thousands of human replays
- Massive computational resources
- Extensive reinforcement learning through self-play

It is far less data-efficient than humans

6. It cannot reason with common sense

AlphaStar has no understanding of:

- Emotions
- Social situations
- Everyday objects
- Human culture

It only understands the game environment in which it was trained.

7. It cannot adapt instantly to major rule changes

If StarCraft II's rules changed significantly, AlphaStar would likely need to be retrained rather than immediately adapting the way an experienced human player could.

8. It cannot operate in the real world by itself

AlphaStar has:

- No ability to perceive the physical world
- No hearing
- No robotic body
- No interaction with real-world objects

It exists only within the StarCraft II environment.

Key takeaway

AlphaStar is one of the most successful examples of **specialized (narrow) AI**. It demonstrated that reinforcement learning can achieve world-class performance in a highly complex strategy game, but it does **not** possess general intelligence or the broad reasoning abilities that humans have.

Capability	Can AlphaStar Do It?
Play StarCraft II	Yes
Beat professional StarCraft II players	Yes
Play any game without retraining	No
Drive a car	No
Hold conversations	No
Explain its reasoning	No
Learn like a human	No
Use common sense	No
Perform everyday tasks	No